

Chapter 02: Real Sequences

1 - Bounded Sequences:

- ① (u_n) bounded below $\Leftrightarrow u_n \geq m; \exists m \in \mathbb{R}$
- ② (u_n) bounded above $\Leftrightarrow u_n \leq \pi; \exists \pi \in \mathbb{R}$
- ③ (u_n) is bounded $\Leftrightarrow |u_n| \leq \pi; \exists \pi \in \mathbb{R}$

2 - Monotone Sequences:

- ① (u_n) is increasing if: $u_{n+1} - u_n \geq 0$
- ② (u_n) is decreasing if: $u_{n+1} - u_n \leq 0$

3 - Convergent Sequences:

- ① (u_n) Convergent if $\lim_{n \rightarrow \infty} u_n = l; l \in \mathbb{R}$.
↳ unique
- ② (u_n) divergent if $\lim_{n \rightarrow \infty} u_n = +\infty$ or $\lim_{n \rightarrow \infty} u_n = -\infty$ or $\lim_{n \rightarrow \infty} u_n = l_1 \neq l_2$

4 - Monotone Convergence Theorems

- ① If $u_n \nearrow$ and $u_n \leq \pi \Rightarrow u_n$ is Convergent
(such that: $\lim_{n \rightarrow \infty} u_n = \sup A$)
 $A = \{u_n; n \in \mathbb{N}\}$
- ② If $u_n \downarrow$ and $u_n \geq m \Rightarrow u_n$ is CV
(s.t. $\lim_{n \rightarrow \infty} u_n = \inf A$)
 $A = \{u_n; n \in \mathbb{N}\}$

5 - Squeeze Lemma:

Let $(u_n), (v_n), (w_n)$ be real seq. s.t.

$$v_n \leq u_n \leq w_n$$

and $\lim_{n \rightarrow \infty} v_n = \lim_{n \rightarrow \infty} w_n = l$

then (u_n) is Convergent seq and

$$\lim_{n \rightarrow \infty} u_n = l$$

6 - Lemma 2:

if $u_n \leq v_n$ and $\lim_{n \rightarrow \infty} u_n = +\infty$

Then: $\lim_{n \rightarrow \infty} v_n = +\infty$

7 - Lemma 3:

Let $(u_n)_{n \in \mathbb{N}}$ and $(v_n)_{n \in \mathbb{N}}$ be CV seq

and $u_n \leq v_n; \forall n \in \mathbb{N}$

Then $\lim_{n \rightarrow \infty} u_n \leq \lim_{n \rightarrow \infty} v_n$

8 - Geometrical Sequence:

Idea: $u_n = a^n; a \in \mathbb{R}; n \in \mathbb{N}$

$a > 1, a < 1, a = 1, a = -1, a = 0$

① If $-1 < a < 1 \rightarrow \lim_{n \rightarrow \infty} u_n = 0 \rightarrow u_n$ CV

② If $a > 1$.
 $\lim_{n \rightarrow \infty} a^n = \lim_{n \rightarrow \infty} e^{n \ln a} = +\infty \rightarrow u_n$ div.

③ If $a < -1$. ; $a = -b; b > 0$

$$\lim_{n \rightarrow \infty} a^n = \lim_{n \rightarrow \infty} (-b)^n = \lim_{n \rightarrow \infty} (-1)^n (b)^n$$

$$u_n \text{ is div. } \Leftrightarrow \begin{cases} \lim_{n \rightarrow \infty} b^n = +\infty \\ \lim_{n \rightarrow \infty} (-1)^n \text{ Doesn't exist} \end{cases}$$

④ If $a = 1 \rightarrow u_n = 1^n$ is CV.

⑤ If $a = -1 \Rightarrow u_n = (-1)^n$ is div.

Idea:

When you find $\cos(n\pi)$ you replace it directly by $(-1)^n$.

9 - Adjacent sequences : متتاليتان متجاورتان
 $(u_n)_{n \in \mathbb{N}}$ and $(v_n)_{n \in \mathbb{N}}$ are adjacent if:

① $(u_n)_{n \in \mathbb{N}}$ is $\nearrow$ and $(v_n)_{n \in \mathbb{N}}$ is $\searrow$

② $\forall n \in \mathbb{N} : u_n \leq v_n$.

③ $\lim_{n \rightarrow +\infty} (u_n - v_n) = 0$.

Thm: $\lim_{n \rightarrow +\infty} v_n = \lim_{n \rightarrow +\infty} u_n$.

10 - Bolzano-Weierstrass Theorem:

The Seq $(u_n)_{n \in \mathbb{N}}$ has a subseq $(c_n)_{n \in \mathbb{N}}$

Such that: $\rightarrow u_n$ is convergent then:
 every subsequence of (u_n) is CV.

Find the subseq of $u_n = (-1)^n$.

$$u_{2n} = 1 \quad \text{and} \quad u_{2n+1} = -1$$

$$c_1(n) = 2n \quad | \quad c_2(n) = 2n+1$$

11 - Recurrent sequence : متتالية عودية

$$\begin{cases} u_0 \in \mathbb{I} \\ u_{n+1} = f(u_n) : n \in \mathbb{N} \end{cases}$$

proposition:

let f be a continuous funct and we have

$\lim_{n \rightarrow +\infty} u_n = l$; then l is a solution of the

equation $f(l) = l$.